

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

HUMAN NUTRITION

(Common to CE, EEE, ME, ECE, CSE, IT, AI&DS, CSE (AI), CSE (DS), CSE (AI&ML), CSE (IoT), CSE(CS), CS&D and AI&ML)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- (a) Name the energy providing constituents in food. 2M
 - (b) Differentiate clearly between essential and non essential amino acids. 2M
 - (c) Name the factors affecting BMR. 2M
 - (d) What do you understand by Balanced diet? 2M
 - (e) Define over nutrition. 2M
 - (f) Define under nutrition. 2M
 - (g) Define Biological value. 2M
 - (h) What do you understand by reference protein in determining chemical score? 2M
 - (i) Name the potentially undesirable constituent in bitter almond. 2M
 - (j) Name the potentially undesirable constituent in cabbage. 2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Explain the process of fat digestion in human nutrition. 10M
- OR**
- 3 What is National Nutritional Policy of Government of India? Explain its main objectives. 10M
- 4 Explain the important factors responsible to determine Recommended Dietary Allowances. 10M
- OR**
- 5 What are the dietary recommendations for sport persons? What food do athletes avoid? 10M
- 6 What is Pellagra disease? Explain its causes. 10M
- OR**
- 7 Write the chemical name, sources and deficiency diseases of the following vitamins: 10M
(i) Thiamine, (ii) Vitamin B₁₂.
- 8 Explain anthropometric method to determine nutritional status. 10M
- OR**
- 9 How NPU is determined, explain clearly? 10M
- 10 Enumerate the composition of blood. Also mention the functions of Plasma. 10M
- OR**
- 11 What is a hormone? What does it work? Name any three diseases caused by hormonal imbalance. Name the hormone produced by the adrenal glands. 10M
